Nautilus Protocol

Technical Specification v0.4

A Fibonacci-based token launch framework for Solana

Fibonacci controls supply. The market controls price. Nobody controls anything else.

Just math.

1. Overview

Nautilus is an open-source token launch framework for Solana. It uses an asymmetric pricing model where buy price follows the Fibonacci sequence and sell price follows a weighted average of treasury balance. The treasury is a program-derived address with no private key — funds move only through the protocol's instructions.

2. Architecture

Component	Implementation	Notes
Buy price	Fibonacci fixed	BASE_PRICE × FIB[stage]
Sell price	Weighted average	treasury_balance ÷ total_sold
Treasury	Program-derived address	No private key exists
Mint authority	Program-derived address	No private key exists
Admin functions	None	No on-chain admin
Upgrade authority	Held by deployer	See Section 8

3. Pricing Mechanism

3.1 Buy Price — Fibonacci Fixed

```
buy_price = BASE_PRICE_LAMPORTS × FIB[current_stage]
BASE_PRICE_LAMPORTS = 1,000,000 (0.001 SOL)
```

The price schedule is immutable after deployment. At SOL ≈ $82, Stage 1 market cap is approximately $82,000 when Stage 1 supply is exhausted.

3.2 Sell Price — Weighted Average

```
sell_price = state.treasury_balance ÷ state.total_sold
```

When total_sold is zero, sell price is undefined. UIs should display base price or N/A.

3.3 Spread

```
payout = sell_price × amount × 0.995
```

The 0.5% spread remains in the treasury. This is the source of the floor price ratchet described in Section 5.

3.4 Price Source of Truth

Sell price is calculated from state.treasury_balance (accounted), not actual PDA lamports. Direct SOL transfers to the treasury PDA do not affect pricing.

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4. Fibonacci Supply Stages

4.1 Stage Table (Stages 1–10)

Stage	Buy price	Sell price*	Buy/Sell ratio	vs Golden ratio φ	Cumulative supply
1	0.0010 SOL	0.0010 SOL	1.0000	φ の 61.8%	1M 枚
2	0.0010 SOL	0.0010 SOL	1.0000	φ の 61.8%	2M 枚
3	0.0020 SOL	0.0015 SOL	1.3333	φ の 82.4%	4M 枚
4	0.0030 SOL	0.0021 SOL	1.4000	φ の 86.5%	7M 枚
5	0.0050 SOL	0.0033 SOL	1.5000	φ の 92.7%	12M 枚
6	0.0080 SOL	0.0052 SOL	1.5385	φ の 95.1%	20M 枚
7	0.0130 SOL	0.0083 SOL	1.5714	φ の 97.1%	33M 枚

8	0.0210 SOL	0.0132 SOL	1.5882	φ の 98.2%	54M 枚
9	0.0340 SOL	0.0213 SOL	1.6000	φ の 98.9%	88M 枚
10	0.0550 SOL	0.0342 SOL	1.6067	φ の 99.3%	143M 枚

* Sell price at stage completion (weighted average of all purchases up to that stage).

4.2 The Golden Ratio Property

As stages progress, the buy/sell ratio converges toward $\varphi \approx 1.618033$. This is a mathematical consequence of the Fibonacci sequence, not an explicit design parameter. At Stage 10 the ratio is 1.6067 (0.7% from φ). At Stage 20 it reaches 1.6179 (0.01% from φ).

4.3 Stage Advancement

Stage advances automatically when supply is exhausted. Advancement is irreversible.

4.4 Maximum Amount Per Transaction

Each instruction is limited to 1,000,000 tokens per transaction to prevent arithmetic overflow at high Fibonacci stages.

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5. Two Core Properties

5.1 The Treasury Cannot Be Drained

The treasury can always fulfill all sell orders.

```
payout          = (T / N) × k × 0.995
treasury_after  = T × (1 - 0.995k/N)
```

As long as k < N, treasury_after > 0. The treasury reaches zero only when the last token is sold. This is a mathematical guarantee, independent of market conditions.

5.2 The Floor Price Almost Never Decreases

After virtually every sell transaction, the sell price (floor price) increases.

Since 0.995k/N < k/N:

```
sell_price_after = T(1 - 0.995k/N) / (N-k) > T/N = sell_price_before
```

The seller receives 99.5% of their proportional share. The remaining 0.5% (spread) stays in treasury, raising the floor for all remaining holders. The more selling occurs, the higher the floor rises for those who remain.

Note: In edge cases involving extremely large single transactions relative to total supply, the floor may marginally decrease. This is practically rare.

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6. Treasury

6.1 Program-Derived Address

```
seeds: [b"treasury", state.key()]
```

No private key exists. SOL can only leave via the sell instruction, which requires burning tokens. The deployer cannot withdraw treasury funds.

6.2 Accounted vs Actual Balance

state.treasury_balance is the source of truth for all price calculations. Direct SOL transfers to the PDA do not affect pricing.

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7. Stage Economics

7.1 Later Participants Raise the Floor

Stage N buyers pay FIB[N] × BASE_PRICE — above the current weighted average. The excess SOL remains in treasury, raising the sell price for all existing holders.

7.2 Profit by Stage (Stage 16 Completion Basis)

Estimated profit multiples assuming SOL = $82 and all stages through Stage 16 sell out. Stage 16 MC ≈ $129B USD. Stage 15 is approximately break-even; Stage 16 results in a loss. No profit is guaranteed at any stage.

Entry stage	Buy price	Entry MC	Multiple	P&L
1	0.0010 SOL	0.00B USD	607.2x	+60618%
2	0.0010 SOL	0.00B USD	607.2x	+60618%
3	0.0020 SOL	0.00B USD	303.6x	+30259%
4	0.0030 SOL	0.00B USD	202.4x	+20139%
5	0.0050 SOL	0.00B USD	121.4x	+12044%
6	0.0080 SOL	0.01B USD	75.9x	+7490%
7	0.0130 SOL	0.02B USD	46.7x	+4571%
8	0.0210 SOL	0.06B USD	28.9x	+2791%
9	0.0340 SOL	0.15B USD	17.9x	+1686%

10	0.0550 SOL	0.40B USD	11.0x	+1004%
11	0.0890 SOL	1.05B USD	6.8x	+582%
12	0.1440 SOL	2.75B USD	4.2x	+322%
13	0.2330 SOL	7.20B USD	2.6x	+161%
14	0.3770 SOL	18.86B USD	1.6x	+61%
15	0.6100 SOL	49.37B USD	1.00x	-0%
16	0.9870 SOL	129.25B USD	0.62x	-38%

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8. Upgrade Authority

Upgrade authority is currently held by the deployer. No on-chain admin functions exist.

Version	Upgrade authority status
v0.4 (current)	Held by deployer
v0.5 (planned)	Revoked — program immutable

```
solana program show <PROGRAM_ID>
```

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9. Implementation

9.1 Instructions

- initialize — creates state account, SPL mint, treasury PDA
- buy(amount) — transfers SOL to treasury, mints tokens to buyer ATA
- sell(amount) — burns tokens, transfers SOL from treasury PDA
- get_state — read-only query of stage, prices, treasury balance

9.2 Tech Stack

- Runtime: Solana / Framework: Anchor 0.32.1 / Token: SPL Token / Language: Rust + TypeScript

9.3 CLI

```
nautilus status  <STATE_ADDRESS>
nautilus buy     <STATE_ADDRESS> <amount>
nautilus sell    <STATE_ADDRESS> <amount>
nautilus balance <STATE_ADDRESS>
```

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10. Known Limitations

- Upgrade authority remains with deployer in v0.4.
- Floor price ratchet may not hold for extremely large single transactions relative to total supply.
- Direct SOL transfers to treasury PDA do not affect pricing.
- Stage supply caps are global, not per-wallet.
- No profit is guaranteed at any stage.
- Maximum 1,000,000 tokens per transaction.
